

Annotated follow-along guide__EDA using basic data functions with Python

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1 Annotated follow-along guide: EDA using basic data functions with Python

This notebook contains the code used in the following instructional video: [EDA using basic data functions with Python](#).

1.1 Introduction

Throughout this notebook, we will implement discovering skills on a dataset. Before getting started, watch the associated instructional video and complete the in-video question. All of the code we will be implementing and related instructions are contained in this notebook.

1.2 Overview

In this notebook, we will use pandas to examine 2018 lightning strike data collected by the National Oceanic and Atmospheric Administration (NOAA). Then, we will calculate the total number of strikes for each month and plot this information on a bar graph.

1.3 Import packages and libraries

Before getting started, we will need to import all the required libraries and extensions. Throughout the course, we will be using pandas, numpy, and datetime for operations, and matplotlib, pyplot, and seaborn for plotting.

```
[1]: import pandas as pd
import numpy as np
import datetime as dt
import matplotlib.pyplot as plt
```

```
[3]: # Read in the 2018 lightning strike dataset.
df = pd.read_csv('eda_using_basic_data_functions_in_python_dataset1.csv')
```

```
[3]: # Inspect the first 10 rows.
df.head(10)
```

```
[3]:      date  number_of_strikes  center_point_geom
0  2018-01-03             194    POINT(-75 27)
1  2018-01-03             41    POINT(-78.4 29)
2  2018-01-03             33    POINT(-73.9 27)
3  2018-01-03             38    POINT(-73.8 27)
4  2018-01-03             92    POINT(-79 28)
5  2018-01-03            119    POINT(-78 28)
6  2018-01-03             35    POINT(-79.3 28)
7  2018-01-03             60    POINT(-79.1 28)
8  2018-01-03             41    POINT(-78.7 28)
9  2018-01-03            119    POINT(-78.6 28)
```

Notice that the data is structured as one row per day along with the geometric location of the strike.

A quick way to determine how many rows and columns of data there are in total is to use `df.shape`. The information will be output as: `([rows], [columns])`.

```
[4]: df.shape
```

```
[4]: (3401012, 3)
```

The total number of rows is 3,401,012, and there are three columns.

```
[5]: # Get more information about the data, including data types of each column
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3401012 entries, 0 to 3401011
Data columns (total 3 columns):
#   Column                Dtype
---  -----
0   date                  object
1   number_of_strikes    int64
2   center_point_geom    object
dtypes: int64(1), object(2)
memory usage: 77.8+ MB
```

1.3.1 Convert the date column to datetime

`info` will provide the total number of rows (3,401,012) and columns (3). It will also state the names and data types of each column, as well as the size of the dataframe in memory.

In this case, notice that the `date` column is an ‘object’ type rather than a ‘date’ type. Objects are strings. When dates are encoded as strings, they cannot be manipulated as easily. Converting string dates to datetime will enable you to work with them much more easily.

Let's convert to datetime using the pandas function `to_datetime()`.

```
[6]: # Convert date column to datetime
df['date'] = pd.to_datetime(df['date'])
```

1.3.2 Calculate the days with the most strikes

As part of discovering, you want to get an idea of the highest data points. For this dataset, we can calculate the top 10 days of 2018 with the most number of lightning strikes using the `groupby()`, `sum()`, and `sort_values()` functions from pandas.

When using `groupby()` on the date column, the function combines all rows with the same date into a single row.

Then, using `sum()` performs a sum calculation on all other summable columns. In this case, we are summing all the lightning strikes that happened on each day. Notice that the `center_point_geom` column is not included in the output. That's because, as a string object, this column is not summable.

Finally, `sort_values()` returns the results in descending order of total strikes for each day in the data.

```
[7]: # Calculate days with most lightning strikes.
df.groupby(['date']).sum().sort_values('number_of_strikes', ascending=False).
    ↪head(10)
```

```
[7]:
```

	number_of_strikes
date	
2018-08-29	1070457
2018-08-17	969774
2018-08-28	917199
2018-08-27	824589
2018-08-30	802170
2018-08-19	786225
2018-08-18	741180
2018-08-16	734475
2018-08-31	723624
2018-08-15	673455

A common mistake among data professionals is using `count()` instead of `sum()`, and vice versa. In this case, `count()` would return the number of occurrences of each date in the dataset, which is not the desired result.

1.3.3 Extract the month data

Next, we will extract the month data from the `date` column and add that extracted month data into a new column called `month`. `dt.month` extracts just the month information (as a numeric value) from the date. This is why converting the `date` column to datetime is very useful.

```
[8]: # Create a new `month` column
df['month'] = df['date'].dt.month
df.head()
```

```
[8]:      date  number_of_strikes  center_point_geom  month
0 2018-01-03             194    POINT(-75 27)         1
1 2018-01-03             41    POINT(-78.4 29)         1
2 2018-01-03             33    POINT(-73.9 27)         1
3 2018-01-03             38    POINT(-73.8 27)         1
4 2018-01-03             92    POINT(-79 28)          1
```

1.3.4 Calculate the number of strikes per month

Now, we will sort our values by most strikes per month. Use `groupby()`, `sum()` and `sort_values()` from pandas again.

```
[9]: # Calculate total number of strikes per month
df.groupby(['month']).sum().sort_values('number_of_strikes', ascending=False).
    ↪head(12)
```

```
[9]:      number_of_strikes
month
8          15525255
7          8320400
6          6445083
5          4166726
9          3018336
2          2071315
4          1524339
10         1093962
1           860045
3           854168
11          409263
12           312097
```

1.3.5 Convert the month number to text

To help read the data more easily, let's convert the month number to text using the datetime function `dt.month_name()` and add this as a new column in the dataframe. `str.slice` will omit the text after the first three letters.

```
[10]: # Create a new `month_txt` column.
df['month_txt'] = df['date'].dt.month_name().str.slice(stop=3)
df.head()
```

```
[10]:      date  number_of_strikes  center_point_geom  month  month_txt
0  2018-01-03             194    POINT(-75 27)        1      Jan
1  2018-01-03             41    POINT(-78.4 29)        1      Jan
2  2018-01-03             33    POINT(-73.9 27)        1      Jan
3  2018-01-03             38    POINT(-73.8 27)        1      Jan
4  2018-01-03             92    POINT(-79 28)         1      Jan
```

1.3.6 Create a new dataframe

The objective is to plot the total number of strikes per month as a bar graph. To help with the plotting, we will create a new dataframe called `df_by_month`. This will allow us to easily access the month, month text, and total number of strikes for each month.

```
[11]: # Create a new helper dataframe for plotting.
df_by_month = df.groupby(['month', 'month_txt']).sum().sort_values('month',
↪ascending=True).head(12).reset_index()
df_by_month
```

```
[11]:      month  month_txt  number_of_strikes
0         1      Jan             860045
1         2      Feb            2071315
2         3      Mar             854168
3         4      Apr            1524339
4         5      May            4166726
5         6      Jun            6445083
6         7      Jul             8320400
7         8      Aug            15525255
8         9      Sep             3018336
9        10      Oct            1093962
10       11      Nov             409263
11       12      Dec             312097
```

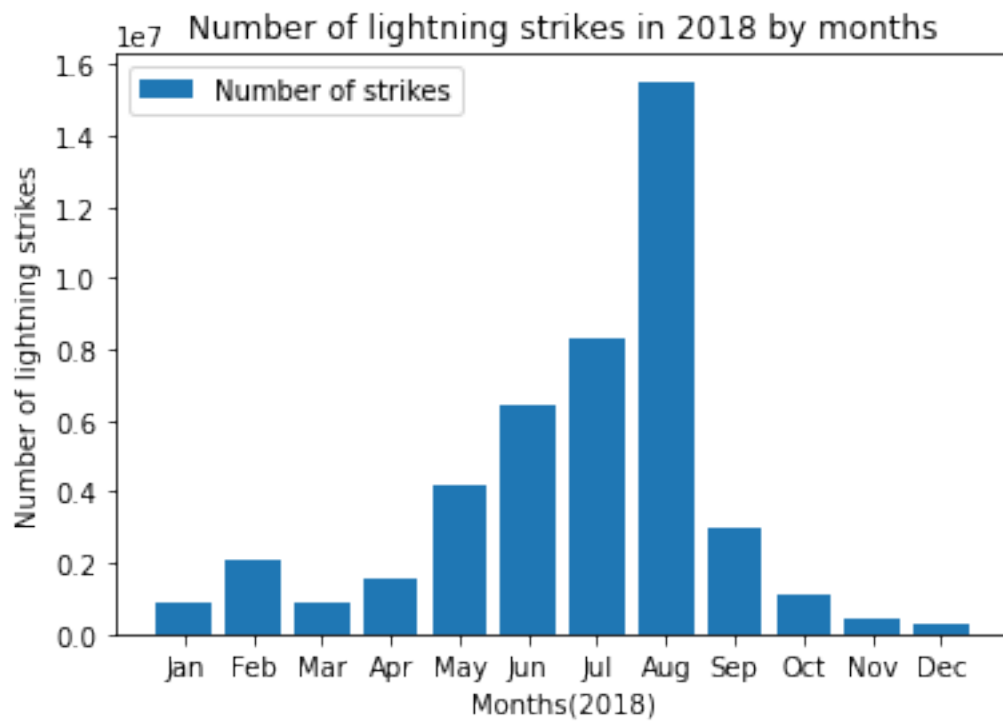
1.3.7 Make a bar chart

Now, let's make a bar chart. Pyplot's `plt.bar()` function takes positional arguments of `x` and `height`, representing the data used for the x- and y- axes, respectively. The x-axis will represent months, and the y-axis will represent strike count.

```
[12]: plt.bar(x=df_by_month['month_txt'], height= df_by_month['number_of_strikes'],
↪label="Number of strikes")
plt.plot()

plt.xlabel("Months(2018)")
plt.ylabel("Number of lightning strikes")
plt.title("Number of lightning strikes in 2018 by months")
```

```
plt.legend()  
plt.show()
```



1.4 Conclusion

If you have successfully completed the material above, congratulations! You now have some of the fundamental elements of data discovery that you can apply to your own datasets.